Data handling & Visualization | Activity 9 | Yveerendra Reddy
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①. Time Series & trend :-

```
n<- length (sales)
fit <- lm (sales~t)
plot(x)
coef (fit)[2]
```

Answer:- n = 44, cycle length = 4 quarters, full cycles = 11, trend is rising, Slope = 3.977 lakh/qua

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②. Seasonal pattern :-

```
d<- decompose (x, type="multiply")
d$figure
which.max (d$ figure)
which.min (d$ figure)
diff (range(d$ figure)).
```

Answer:- Peak = Q4, trough = Q2 - Theoretical Seasonal factors are : 1.0, 0.7, 1.0, 1.3; theorethical swing = 0.60.

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③. Consistency:-

```
cy <- matrix (sales, row w=4)
sum (apply (cy, 2, which.max)==4)
matplot (1:4, cy, type = "l", lty=1)
```

Answer:- Seasonality repeats every 4-quarters. The code counts how many of 11 cycles peak in Q4

④. Auto Correlation :-

```
e <- resid (lm (sales ~t))
a <- acf( e, lag.max = 4, plot = FALSE )$acf
a [c (3,5)]
2/ sqrt(44)
```

Answer:- Significance bound = 0.302, Compare absolute ACF values at lags 2 & 4 with 0.302.

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⑤. Seasonality type:-

```
cy <- matrix (sales, nrow=4)
r1 <- differ (range((cy[,1]))
r2 <- diff (range (cy[,11]))

c (r1, r2, r2/r1)
adj <- sales / rep (d$figure, 11)
coef (lm (adj ~t)) [2]
```

Answer:- Multiplicative seasonality, Sales are expected to peak in Q4 & reach a trough in Q2. plan inventory & staffing accordingly.